

Are DNA-damaging effects induced by herbicide formulations (Roundup® and Garlon®) in fish transient and reversible upon cessation of exposure?



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ABSTRACT

Owing to the seasonality of crop cultivation and subsequent periodic/seasonal application of herbicides, their input to the aquatic systems is typically intermittent. Consequently, exposure of fish to this type of contaminants can be short and followed by a period of permanence in non-contaminated areas. Thus, the assessment of genotoxic endpoints in fish after removal of the contamination source appears as a crucial step to improve the knowledge on the dynamics of herbicide genotoxicity, as well as to determine the actual magnitude of risk posed by these agrochemicals. Therefore, the present study intended to shed light on the ability of fish to recover from the DNA damage induced by short-term exposures to the herbicide formulations Roundup® (glyphosate-based) and Garlon® (triclopyr-based) upon the exposure cessation. European eel (*Anguilla anguilla*) was exposed to the above commercial formulations for 3 days, and allowed to recover for 1, 7 and 14 days (post-exposure period). The comet assay was used to identify the DNA damage in blood cells during both exposure and post-exposure periods. As an attempt to clarify the DNA damaging mechanisms involved, an extra-step including the incubation of the nucleotides with DNA lesion-specific repair enzyme was added to the standard comet. The genotoxic potential of both herbicides was confirmed, concerning the exposure period. In addition, the involvement of oxidative DNA damage on the action of Roundup® (pointed out as pyrimidine bases oxidation) was demonstrated, while for Garlon® this damaging mechanism was less evident. Fish exposed to Garlon®, though presenting some evidence towards a tendency of recovery, did not achieve a complete restoration of DNA integrity. In what concerns to Roundup®, a recovery was evident when considering non-specific DNA damage on day 14 post-exposure. In addition, this herbicide was able to induce a late oxidative DNA damage (day 14). Blood cells of *A. anguilla* exposed to Roundup® appeared to be more successful in repairing damage with a non-specific cause than that associated to base oxidation. Overall, the present findings highlighted the genetic hazard to fish associated to the addressed agrochemicals, reinforcing the hypothesis of long-lasting damage.

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1. Introduction

Aquatic contamination became a huge environmental concern since the aquatic compartment was converted into a receptacle of several undesirable contaminants, such as agrochemicals. In this context, herbicides, which are widely used to control weeds in forestry and agriculture, can reach the aquatic systems by uncontrolled runoff, aerial drift, or inadvertent overspray, affecting fish in particular. Owing to the seasonality of crops and subsequent

periodic/seasonal application of these weed-killers, their input to the aquatic systems is typically intermittent. For this reason, as well as due to fish behaviour (e.g. avoidance responses) and mobility (e.g. feeding, refuge-seeking or spawning migrations), their exposure to this type of contaminants can be short (time-scale of days) and followed by a period of permanence in non-contaminated areas. Therefore, the assessment of toxicity endpoints in fish after removal of the contamination source, seeking for an eventual recovery from damage previously induced appears as a crucial step to improve the knowledge on the dynamics of herbicide-induced toxicity, as well as to determine the actual magnitude of risk posed by these agrochemicals to fish. Although several ecotoxicological studies have been performed addressing the effects exerted by

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herbicides in fish (Crestani et al., 2007; Moraes et al., 2007; da Fonseca et al., 2008), the evaluation of the same effects in the post-exposure period was only described by one study (Menezes et al., 2011).

Due to role of DNA as repository of genetic information, its integrity and stability are critical factors to life. Therefore, the evaluation of herbicides as potential exogenous sources of DNA damage to fish is a matter that deserves scientific efforts towards a deepen knowledge on the risks associated with these contaminants. Moreover, the perspective previously presented, related to the (hypothetical) transient or reversible nature of the effects, applied to herbicide-induced DNA lesions is completely absent in the literature.

As defined for the majority of toxic effects, genotoxicity can be determined by a cascade of events, beginning with exposure and culminating with the expression of one or more endpoints. This cascade includes genotoxicant absorption, distribution, metabolism (both detoxification and activation), distribution of metabolites, and interaction with nucleic acids (DNA and RNA), damage repair and, finally, excretion (Hodgson, 2012). Hence, the progression of genetic lesions in a post-exposure period is determined by the combined interference of the mentioned processes, with DNA repair playing a central role. As a general knowledge, it is recognized that genetic damage could be originated by DNA adduct formation, DNA/DNA, DNA/protein cross-links as well as DNA single/double strand-breaks (e.g. as a consequence of base oxidation) and their repair. Therefore, DNA repair processes, though essential for cell viability, can play an ambivalent role, diminishing the impact of the genotoxicant on one hand and, on the other, unintentionally amplifying that impact. Several mechanisms of repair can be considered. The reversal of damage is the most direct repair mechanism and the most efficient. When the damage is more severe, the repair required may presume the excision of a base (base excision repair—BER) or even a nucleotide (nucleotide excision repair—NER) (Costa and Eaton, 2006). In addition, the mismatch repair (MMR) usually occurs as a result of a replication error in which DNA polymerase incorporates the wrong base (Costa and Eaton, 2006).

Popular herbicide formulations, such as Roundup® and Garlon®, are widely used to for site preparation and conifer release in forestry (Relyea, 2005; Wojtaszek et al., 2005). Although used with similar purposes, these herbicides belong to different chemical classes. Roundup® is an organophosphate glyphosate-based herbicide, while Garlon® has triclopyr acid as its active ingredient, corresponding to the class of pyridinecarboxylic acids. Despite the need of more studies in order to clarify the processes involved in their damaging action, the genotoxic potential of Roundup® and Garlon® was already demonstrated in fish, following short-term exposures (Çavas and Könen, 2007; Cavalcante et al., 2008; Guilherme et al., 2012b, 2014).

Taking as the point of departure the genotoxic potential of Roundup® and Garlon®, considering environmentally realistic concentrations, the present study intended to shed a light on the ability of fish to recover from the DNA damage exerted by short-term exposures to these herbicide formulations upon the exposure cessation, thereby contributing to a realistic perspective of the risk assessment. Hence, European eel (*Anguilla anguilla* L.) was exposed to the previously mentioned commercial formulations for 3 days, and allowed to recover in herbicide-free water for 1, 7 and 14 days (post-exposure period). The comet assay was used to identify DNA damage in blood cells at each moment, during the exposure and post-exposure periods. As an attempt to clarify the involved DNA damaging mechanisms, besides the standard procedure, comet assay was carried out including an extra step where nucleoids were incubated with DNA lesion-specific bacterial repair enzymes (Endonuclease III—EndoIII and formamidopyrimidine DNA glycosylase—FPG) which convert base damage into

breaks, signaling oxidized pyrimidines and purines, respectively (Azqueta et al., 2009). Thus, this methodology allows the detection of genotoxic risk resulting from unspecific (alkali-labile sites and single strand breaks associated with incomplete excision repair sites) as well as specific (bases oxidation) damage. Hence, this additional step intends to clarify if the induced damage has an oxidative cause, as well as, in case of an eventual recovery, to relate the recovery rates with the type of damage.

2. Material and methods

2.1. Chemicals

The experiment was conducted using two commercial formulations: Roundup® Ultra, distributed by Bayer CropScience (Portugal), containing isopropylammonium salt of glyphosate at 485 g L^{-1} as the active ingredient (equivalent to 360 g L^{-1} or 30.8% w/v of glyphosate) and POEA (16% v/v) as surfactant, and Garlon®, distributed by Dow AgroSciences (Lusosem, Portugal), containing triclopyr formulated as a butoxyethyl ester, at a concentration of 480 g L^{-1} (or 44.4% w/v) as well as kerosene (petroleum distillate) as an adjuvant. DNA lesion-specific repair enzymes, namely formamidopyrimidine DNA glycosylase (FPG) and endonuclease III (EndoIII) were purchased from Professor Andrew Collins (University of Oslo, Norway). All the other chemicals needed to perform comet assay were obtained from the Sigma-Aldrich Chemical Company (Spain).

2.2. Test animals and experimental design

European eel (*Anguilla anguilla* L.) juvenile specimens (glass eel stage) were captured from Minho river mouth, Caminha, Portugal. Eels were acclimatized to laboratory for 20 days and kept in 20-L aquaria under natural photoperiod in aerated, filtered, dechlorinated and recirculating tap water with the following physico-chemical conditions: salinity 0, temperature $20 \pm 1^\circ \text{C}$, pH 7.1 ± 0.3 , nitrate $27 \pm 0.2 \text{ mg L}^{-1}$, nitrite $0.07 \pm 0.02 \text{ mg L}^{-1}$, ammonia $0.2 \pm 0.04 \text{ mg L}^{-1}$, dissolved oxygen $8.2 \pm 0.3 \text{ mg L}^{-1}$. During this period, fish were fed with fish roe daily. The experiments were carried out in 1-L aquaria, in a semi-static mode, under the conditions described for the acclimation period. After acclimation, eels (weighing $0.25 \pm 0.02 \text{ g}$, with total length of $7.1 \pm 0.2 \text{ cm}$) were exposed to $270.5 \mu\text{g L}^{-1}$ of Garlon® (G group) for 3 days (exposure period). Another set of fish was exposed for 3 days to the previously mentioned Garlon® concentration and thereafter transferred to herbicide-free water and sampled after 1, 7 and 14 days (post-exposure period). In parallel, groups of fish were kept in herbicide-free water and sampled at the same times (control groups). The experiment (exposure and post-exposure periods) was carried out using aquaria triplicates ($n=3$) where each aquarium had a group of 4 fish ($3 \times 4 = 12$ fish for each condition/time). Water medium in both exposure and post-exposure periods was daily renewed (100%).

Simultaneously, the same experimental design was repeated, this time exposing fish to $116 \mu\text{g L}^{-1}$ Roundup® (R group).

Tested concentrations of both commercial formulations were based on previous studies performed by the authors (Guilherme et al., 2010, 2012a, 2012b, 2014), where these concentrations were demonstrated to be able of exerting genotoxicity in fish.

Fish were not fed during the exposure period, but daily fed with fish roe during the post-exposure period. Fish were sacrificed by cervical transection at the post-opercular region and blood collected from the heart using heparinised capillary tubes. Two μL of blood were immediately diluted in 1 mL of ice-cold phosphate-buffered saline (PBS) to prepare a cell suspension, which was kept on ice until further treatment.

2.3. Evaluation of genetic damage

The conventional alkaline version of the comet assay was performed according to the methodology of Collins (2004) as adapted by Guilherme et al. (2010), with the proper adjustments to assay procedure with an extra step of digesting the nucleoids with endonucleases. A system of eight gels per slide was adopted, based on a model created by Shaposhnikov et al. (2010), in order to increase the assay output. Briefly, 20 μ L of cell suspension (previously prepared in PBS) were mixed with 70 μ L of 1% low melting point agarose, in distilled water. Eight drops with 6 μ L of cell suspension were placed onto the precoated slide (with 1% normal melting point agarose) as two rows of 4 (4 groups of 2 replicates), without coverslips, containing each gel approximately 1500 cells. The gels were left for ± 5 min at 4 °C in order to solidify agarose, and then immersed in a lysis solution (2.5 M NaCl, 0.1 M EDTA, 10 mM Tris, 1% Triton X-100, pH 10) at 4 °C, for 1 h. After lysis of agarose-embedded cells, slides were washed 3 times with buffer (0.1 M KCl, 0.5 mM EDTA, 40 mM HEPES, 0.2 mg mL⁻¹ bovine serum albumin, pH 8) at 4 °C.

Three sets of slides were prepared: two sets were incubated with endonucleases FPG or EndoIII, which convert oxidized purines and pyrimidines into DNA single strand breaks, respectively (Azqueta et al., 2009), and a third set was incubated only with buffer. Hence, 30 μ L of each enzyme (diluted in buffer) were applied in each gel, together with a coverslip, prior to incubation at 37 °C for 30 min in a humidified atmosphere. The slides were then gently placed in the electrophoresis tank, immersed in electrophoresis solution (20 min) for alkaline treatment. DNA was allowed to migrate at a fixed voltage of 25 V, a current of 300 mA which results in 0.7 V cm⁻¹ (achieved by adjusting the buffer volume in the electrophoresis tank). The slides were stained with ethidium bromide (20 μ g mL⁻¹).

Fifty nucleoids were observed per gel, using a Leica DMLS fluorescence microscope (400 $\times$ magnification). The DNA damage was quantified by visual classification of nucleoids into five comet classes, according to the tail intensity and length, from 0 (no tail) to 4 (almost all DNA in tail) (Collins, 2004). The total score expressed as a genetic damage indicator (GDI) was calculated multiplying the percentage of nucleoids in each class by the corresponding factor, according to this formula:

$$\begin{aligned} \text{GDI} = & [(\% \text{ nucleoids class } 0) \times 0] + [(\% \text{ nucleoids class } 1) \times 1] \\ & + [(\% \text{ nucleoids class } 2) \times 2] \\ & + [(\% \text{ nucleoids class } 3) \times 3] \\ & + [(\% \text{ nucleoids class } 4) \times 4] \end{aligned}$$

GDI values were expressed as arbitrary units in a scale of 0 to 400 per 100 scored nucleoids (as average value for the 2 gels observed per fish). When the comet assay was performed with additional FPG and EndoIII steps, GDI values were calculated in the same way but the parameter designated GDI_{FPG} and GDI_{EndoIII}, respectively. Additional DNA breaks corresponding to net enzyme-sensitive sites alone (NSS_{FPG} or NSS_{EndoIII}) were also expressed. Moreover, the frequency of nucleoids observed in each comet class considering GDI_{FPG} and GDI_{EndoIII} was also expressed, as recommended by Azqueta et al. (2009).

2.4. Statistical analysis

Statistica 7.0 software was used for statistical analysis. All data were first tested for normality and homogeneity of variance to meet statistical demands. One-way Analysis of Variance (ANOVA), followed by Dunnett test as Post-hoc comparison, was applied to

compare the treated groups with the control, within the same exposure duration. Concerning the post-exposure period, a two-way ANOVA was applied to test the effect of the factors “agent” and “exposure time” on the levels of DNA damage, as well as the interactions between them. The Tukey test was applied as Post-hoc comparison. In all the analyses, differences between means were considered significant when $p < 0.05$ (Zar, 1996).

3. Results

3.1. Garlon® experiment

3.1.1. Non-specific DNA damage

The analysis of the exposure period revealed a significant increase of the damage exerted by Garlon®, when compared to the control group (Fig. 1A). The post-exposure period (Fig. 1A) showed that fish previously treated with Garlon® still express significantly higher damage values than the respective controls for all sampling times. Moreover, the G group relative to 14 days showed a significant decrease in relation to the other G groups (1 and 7 days post-exposure). Significant effects of the factors agent and time were observed (Table 1) (2-way ANOVA results).

3.1.2. Specific oxidative DNA damage

The detection of oxidized bases was achieved by the comet assay with an extra step where nucleoids were incubated with the DNA lesion-specific repair enzymes FPG and EndoIII (Fig. 1B–E).

3.1.2.1. FPG associated damage. Similarly to what was observed for GDI in the exposure period, the GDI_{FPG} parameter revealed a significant increase of damage for G group, in comparison with control (Fig. 1B). The post-exposure period (Fig. 1B) displayed this kind of difference only between groups relative to 1 and 7 days. The comparison between groups previously exposed to Garlon® revealed a significant damage decrease after 14 days when compared to 1 and 7 days post-exposure. The NSS_{FPG} parameter did not display any significant variation, either considering the exposure or the post-exposure periods (Fig. 1C).

Table 1 revealed, for GDI_{FPG}, a significant effect of agent and time, as well as a significant interaction between them (agent $\times$ time). NSS_{FPG} showed to be unaffected by these factors.

The DNA damage classes were individually analyzed concerning the GDI_{FPG} parameter (Table 2). The exposure period showed significant differences concerning all classes (excluding class 0), in relation to control. Moreover, the control group displayed class 2 as the most frequent, while G group presented a higher number of nucleoids corresponding to class 3. Looking to the post-exposure period, it was evident a significant increase of class 4 for group G, in relation to control, concerning the day 1. The day 7 displayed a significant decrease in class 1 while an increase was detected for class 4, when G group was compared to control. In the 14th day, G group presented a general decrease of the damage in relation to 1 and 7 days, reflected in the increase of class 1 frequency, in parallel with the decreases of classes 3 and 4. At this point in time, class 2 was found as the predominant in both control and G groups.

3.1.2.2. EndoIII associated damage. In accordance with GDI and GDI_{FPG}, the GDI_{EndoIII} parameter (Fig. 1D) pointed G group as presenting significantly higher damage than control, when the exposure period was considered. Concerning the post-exposure period (Fig. 1D), G groups corresponding to 1 and 14 days kept damage levels significantly higher than the respective controls. In line with what was described to the NSS_{FPG}, NSS_{EndoIII} parameter (Fig. 1E) was unable to distinguish test groups at any time.

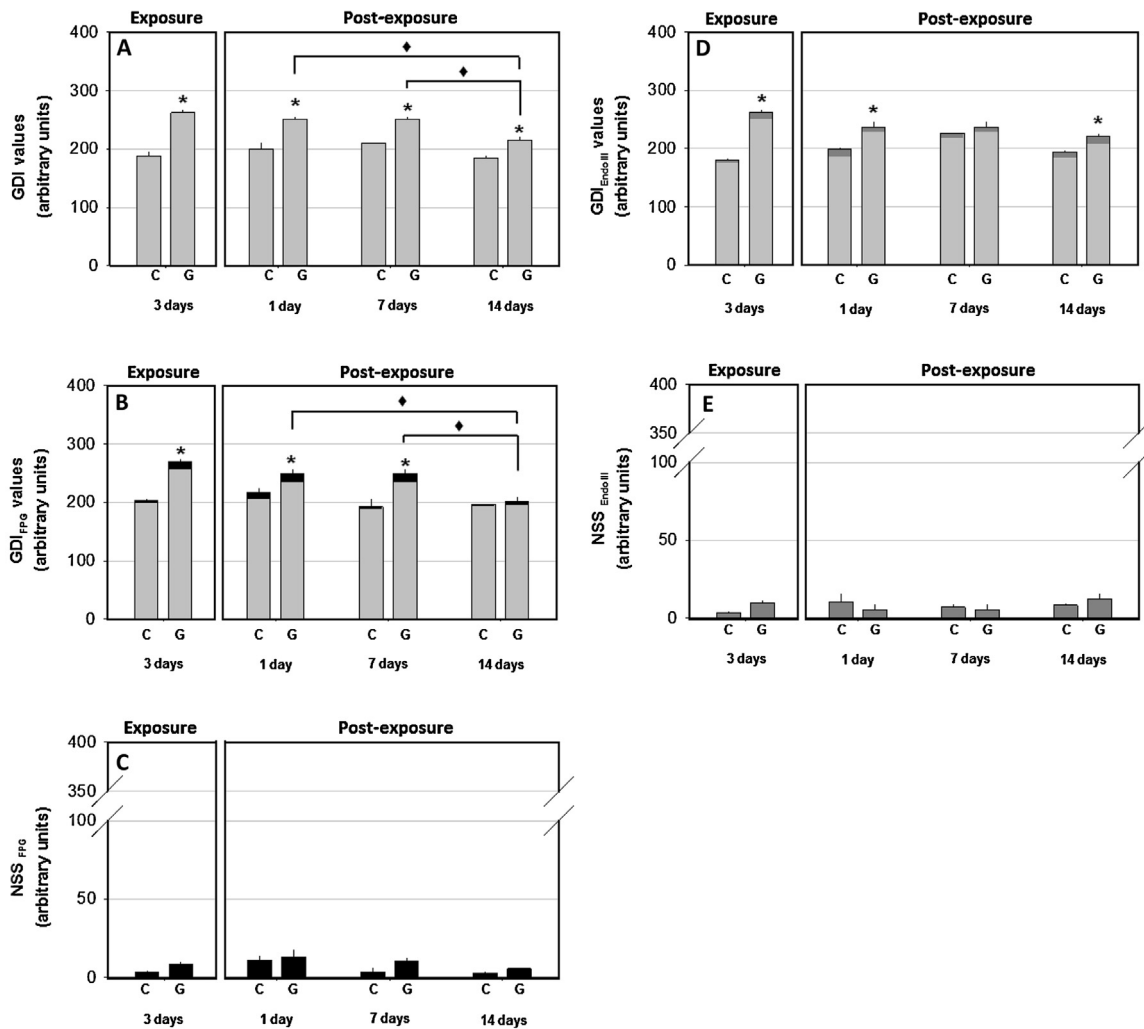


Fig. 1. Mean values of DNA damage, measured by comet assay in blood cells of *A. anguilla* exposed to $270.5 \mu\text{g L}^{-1}$ Garlon® for 3 days (Exposure) and submitted to a period in herbicide-free water for 1, 7 and 14 days (post-exposure); (A) genetic damage indicator (GDI) measured by the standard (alkaline) comet assay; (B) overall damage (GDI_{FPG}) and partial scores, namely genetic damage indicator (GDI; grey) and additional DNA breaks corresponding to net FPG-sensitive sites (NSS_{FPG}; black); (C) NSS_{FPG} alone; (D) overall damage (GDI_{EndoIII}) and partial scores, namely genetic damage indicator (GDI; light grey) and additional DNA breaks corresponding to net EndoIII-sensitive sites (NSS_{EndoIII}; dark grey); (E) NSS_{EndoIII} alone. Bars represent the standard error. Statistically significant differences ($p < 0.05$) are: (*) in relation to control (C), within the same exposure time; (◆) between exposure times, within the same treatment.

Factors as agent and time, as well their interaction (agent $\times$ time) had a significant effect on GDI_{EndoIII} (Table 1). On the other hand, NSS_{EndoIII} did not display any significance.

Looking at the individual analysis of DNA damage classes during the exposure period, and concerning the GDI_{EndoIII} parameter (Table 3), it was possible to notice that G group expressed a significant decrease in classes 1 and 2, while class 3 showed a significant increase, when compared to control. The control group showed class 2 as predominant and G group presented class 3 as the most frequent. The post-exposure period did not present significant

alterations. Moreover, both control and G groups pointed class 2 as the predominant, with the exception of G group, in day 7.

3.2. Roundup® experiment

3.2.1. Non-specific DNA damage

Analysing the exposure period, it was possible to notice that the DNA damage exerted by Roundup®, reflected by the GDI parameter (Fig. 2A), had significantly increased when compared to the control group. Considering the post-exposure period (Fig. 2A), only the R

Table 1
Results of two-way ANOVA testing the effect of agent and time, as well as the interaction between them (agent $\times$ time) on the levels of DNA damage in blood cells of *A. anguilla* in a post-exposure period (1, 7 and 14 days), following a 3 days exposure to $270.5 \mu\text{g L}^{-1}$ Garlon®. Both F and p values are given for each variable. Non significant differences are signalized as “ns”.

Parameter	Factors				Interaction	
	Agent		Time		Agent $\times$ time	
	F	p	F	p	F	p
GDI	92.33	<0.05	20.94	<0.05	2.15	ns
GDI _{FPG}	33.14	<0.05	16.20	<0.05	7.67	<0.05
NSS _{FPG}	1.53	ns	1.16	ns	0.31	ns
GDI _{EndoIII}	34.21	<0.05	10.49	<0.05	3.51	<0.05
NSS _{EndoIII}	0.12	ns	0.68	ns	1.01	ns

Table 2

Mean frequencies (%) of damaged nucleoid classes ($\pm$ standard error), measured by the comet assay including the incubation with the FPG enzyme in blood cells of *A. anguilla* exposed to 270.5 $\mu\text{g L}^{-1}$ Garlon® for 3 days (Exposure) and after cessation of exposure (1, 7 and 14 days post-exposure). Statistically significant differences ($p < 0.05$) are: (°) in relation to control (C), (♦) in relation to 1 day post-exposure and (▲) in relation to 7 days post-exposure.

			GDI _{FPG} —DNA damage Classes—Garlon® experiment				
			0	1	2	3	4
Exposure	3 Days	C	0.00 $\pm$ 0.00	12.25 $\pm$ 1.84	67.50 $\pm$ 1.22	20.25 $\pm$ 0.61	0.00 $\pm$ 0.00
		G	0.00 $\pm$ 0.00	0.42 $\pm$ 0.22°	36.08 $\pm$ 2.62°	57.67 $\pm$ 1.59°	5.83 $\pm$ 1.26°
Post-exposure	1 Days	C	0.00 $\pm$ 0.00	8.00 $\pm$ 3.27	66.92 $\pm$ 1.29	24.83 $\pm$ 4.76	0.25 $\pm$ 0.20
		G	0.00 $\pm$ 0.00	0.58 $\pm$ 0.08	55.00 $\pm$ 4.80	43.67 $\pm$ 5.22	2.00 $\pm$ 0.25°
	7 Days	C	0.00 $\pm$ 0.00	14.75 $\pm$ 0.61	55.83 $\pm$ 0.95	27.42 $\pm$ 1.56	2.00 $\pm$ 0.00
		G	0.00 $\pm$ 0.00	1.00 $\pm$ 0.52°	48.17 $\pm$ 3.25	45.17 $\pm$ 2.57	5.67 $\pm$ 0.36♦
	14 Days	C	0.00 $\pm$ 0.00	12.42 $\pm$ 0.88	74.75 $\pm$ 1.02	12.83 $\pm$ 0.14	0.00 $\pm$ 0.00
		G	0.00 $\pm$ 0.00	18.14 $\pm$ 2.13♦▲	62.08 $\pm$ 3.18	19.44 $\pm$ 4.95♦▲	0.33 $\pm$ 0.17♦▲

groups corresponding to days 1 and 7 post-exposure showed significant differences in relation to their respective control. Moreover, it was noticed that the group which remained 14 days in herbicide-free water revealed a significant damage decrease when compared to the other two R groups (1 and 7 days post-exposure).

Table 4 (2-way ANOVA results) revealed a significant effect of the factors agent and time on GDI levels, as well as a significant interaction agent $\times$ time.

3.2.2. Specific oxidative DNA damage

The detection of oxidized bases was achieved by the comet assay with an extra step where nucleoids were incubated with the DNA lesion-specific repair enzymes FPG and EndoIII (Fig. 2B–E).

3.2.2.1. FPG associated damage. The GDI_{FPG} parameter (Fig. 2B) exhibited an increase of the DNA damage induced by Roundup®, after the exposure period (3 days). When fish previously exposed to Roundup® were transferred to herbicide-free water during 1, 7 and 14 days (Fig. 2B), significant increases were also perceptible for all R groups comparing to the respective control groups. However, the R group relative to 1 day post-exposure displayed significantly higher values when compared with R groups corresponding to 7 and 14 days post-exposure. On the other hand, the NSS_{FPG} parameter (Fig. 2C) only demonstrated significant differences in the post-exposure period. Thus, a significant increase was found comparing R and control groups at day 14. Moreover, at this time, the R group showed a significant NSS_{FPG} increase in comparison to the corresponding groups at 1 and 7 days post-exposure.

Table 4 revealed, for GDI_{FPG}, a significant effect of agent and time. NSS_{FPG} was significantly affected by time and displayed a significant interaction agent $\times$ time.

The damage classes were also analyzed individually, considering the GDI_{FPG} parameter (Table 5). During the exposure period, classes 1 and 2 displayed a decrease, while classes 3 and 4 showed higher values in relation to control. Moreover, it was observed that R group showed the prevalence of class 3, while control group displayed

class 2 with the highest frequency. After the first post-exposure day, the R group displayed a significant decrease in class 2, while classes 3 and 4 assumed values higher than control. The comparison between 1- and 7-day post-exposure showed a significant increase concerning class 2 and displayed lower values for class 3. The fourteenth day presented a significant decrease between control and R groups, concerning the class 2. Moreover, all the classes (once again with the exception of class 0) concerning R group showed to be significantly different from the ones corresponding to the first day. Increases were perceptible considering classes 1 and 2, while classes 3 and 4 presented lower values. Considering the post-exposure period, class 2 was the most frequent with the exception of R group, in day 1.

3.2.2.2. EndoIII associated damage. Considering the exposure period to Roundup®, the GDI_{EndoIII} parameter (Fig. 2D) displayed a significant increase for R group, in comparison with control. In what concerns to the post-exposure period (Fig. 2D), this significant increase was only detected at 1 and 14 days. Moreover, the R group corresponding to day 1 post-exposure assumed significantly higher values than the R groups at 7 and 14 days. The NSS_{EndoIII} parameter (Fig. 2E) displayed a significant damage increase for the R group, when compared with control, considering the exposure period. The post-exposure period (Fig. 2E) revealed exactly the same pattern already described for NSS_{FPG} parameter.

Table 4 revealed, for GDI_{EndoIII}, a significant effect of agent and time, as well as a significant interaction agent $\times$ time. NSS_{EndoIII} was significantly affected by time and displayed a significant interaction agent $\times$ time.

The analysis of individual damage classes in the exposure period, considering the GDI_{EndoIII} parameter (Table 6), revealed the same pattern as observed for GDI_{FPG}. After the first day of post-exposure, differences between control and R groups revealed a decrease in class 1, while class 3 presented higher values. The fourteenth day R group displayed a significant increase in class 1, as well as a significant decrease concerning class 3, when compared to its

Table 3

Mean frequencies (%) of damaged nucleoid classes ($\pm$ standard error), measured by the comet assay including the incubation with the EndoIII enzyme in blood cells of *A. anguilla* exposed to 270.5 $\mu\text{g L}^{-1}$ Garlon® for 3 days (exposure) and after cessation of exposure (1, 7 and 14 days post-exposure). Statistically significant differences ($p < 0.05$) are: (°) in relation to control (C) and (▲) in relation to 7 days post-exposure.

			GDI _{EndoIII} —DNA damage classes—Garlon® experiment				
			0	1	2	3	4
Exposure	3 Days	c	0.00 $\pm$ 0.00	24.25 $\pm$ 1.02	66.75 $\pm$ 0.61	9.00 $\pm$ 0.41	0.00 $\pm$ 0.00
		G	0.00 $\pm$ 0.00	0.92 $\pm$ 0.92°	40.58 $\pm$ 3.25°	53.42 $\pm$ 3.60°	5.08 $\pm$ 1.54
Post-exposure	1 Day	C	0.00 $\pm$ 0.00	19.75 $\pm$ 0.20	61.25 $\pm$ 1.84	19.00 $\pm$ 2.04	0.00 $\pm$ 0.00
		G	0.17 $\pm$ 0.17	6.17 $\pm$ 3.55	52.08 $\pm$ 2.24	40.08 $\pm$ 4.26	1.50 $\pm$ 0.66
	7 Days	C	0.00 $\pm$ 0.00	5.00 $\pm$ 0.82	53.25 $\pm$ 7.55	40.42 $\pm$ 7.28	1.33 $\pm$ 1.09
		G	0.00 $\pm$ 0.00	2.36 $\pm$ 1.50	45.03 $\pm$ 4.47	50.72 $\pm$ 5.13	1.89 $\pm$ 0.75
	14 Days	C	0.00 $\pm$ 0.00	14.25 $\pm$ 5.10	72.67 $\pm$ 4.63	13.08 $\pm$ 0.48	0.00 $\pm$ 0.00
		G	0.00 $\pm$ 0.00	10.31 $\pm$ 0.39	61.14 $\pm$ 3.98	27.31 $\pm$ 3.52▲	1.25 $\pm$ 0.88

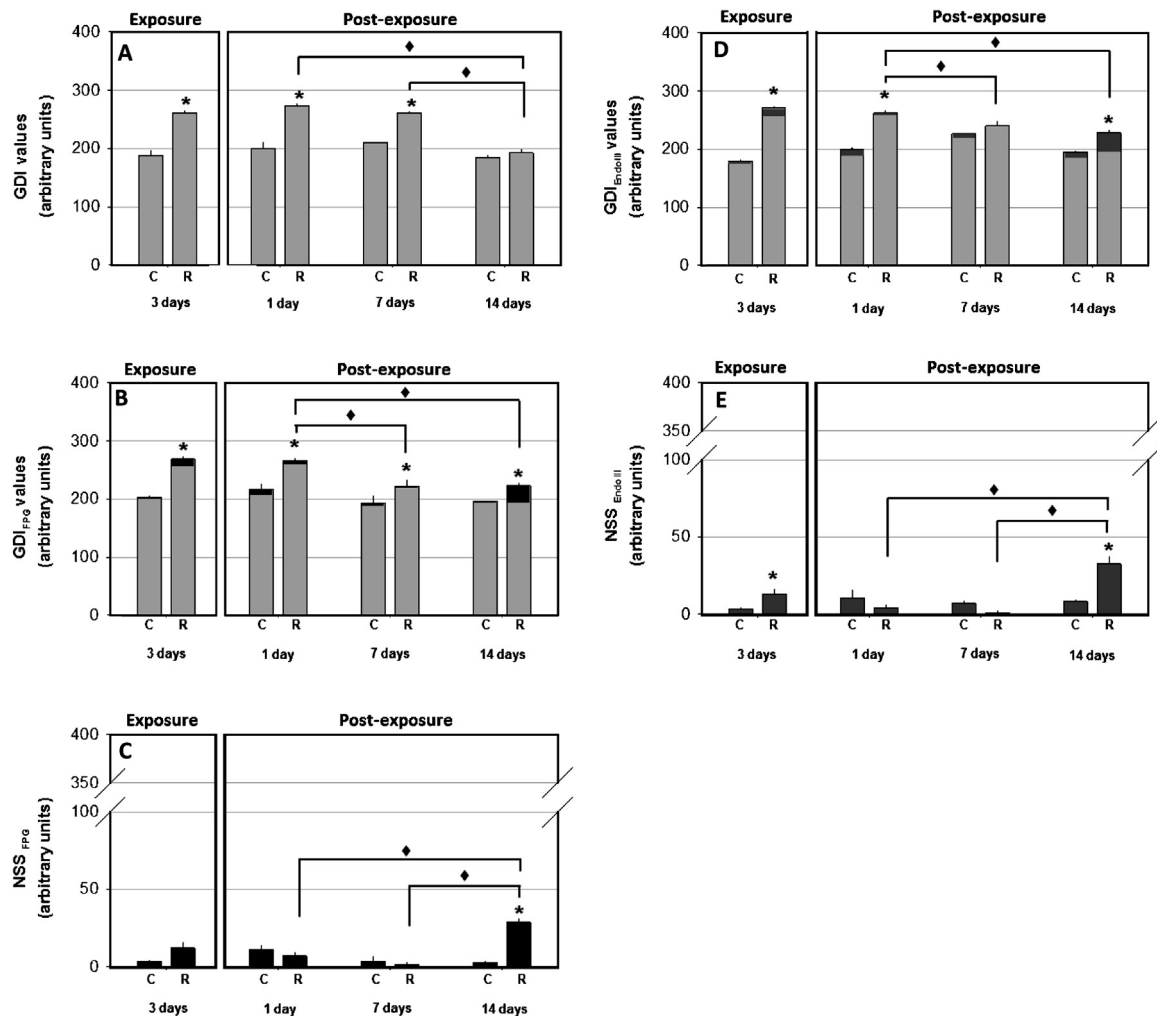


Fig. 2. Mean values of DNA damage, measured by comet assay in blood cells of *A. anguilla* exposed to $116 \mu\text{g L}^{-1}$ Roundup® for 3 days (exposure) and submitted to a period in herbicide-free water for 1, 7 and 14 days (post-exposure); (A) genetic damage indicator (GDI) measured by the standard (alkaline) comet assay; (B) overall damage (GDI_{FPG}) and partial scores, namely genetic damage indicator (GDI; grey) and additional DNA breaks corresponding to net FPG-sensitive sites (NSS_{FPG} ; black); (C) NSS_{FPG} alone; (D) overall damage ($\text{GDI}_{\text{EndoIII}}$) and partial scores, namely genetic damage indicator (GDI; light grey) and additional DNA breaks corresponding to net EndoIII-sensitive sites ($\text{NSS}_{\text{EndoIII}}$; dark grey); (E) $\text{NSS}_{\text{EndoIII}}$ alone. Bars represent the standard error. Statistically significant differences ($p < 0.05$) are: (*) in relation to control (C), within the same exposure time; (◆) between exposure times, within the same treatment.

homologous in day 1. As observed for GDI_{FPG} , class 2 was presented as the prevalent for both groups, with the only exception for R group, in day 1.

4. Discussion

The herbicides tested in the present study, though not extensively studied in terms of their genotoxic potential, have already been considered as DNA damaging agents to fish. Considering this

fact as a starting point, this study aimed to evaluate the DNA damage progression along a post-exposure period. In a recent study of our research group, fish exposed to an environmentally relevant concentration of Garlon® ($270.5 \mu\text{g L}^{-1}$) showed increased DNA damage assessed by the comet assay (Guilherme et al., 2014). However, when the additional breaks corresponding to net enzyme-sensitive sites were considered, no significant levels of oxidative damage were detected (Guilherme et al., 2014). Concerning Roundup®, its genotoxicity to fish has been demonstrated by

Table 4
Results of two-way ANOVA testing the effect of agent and time, as well as the interaction between them (agent $\times$ time) on the levels of DNA damage in blood cells of *A. anguilla* in a post-exposure period (1, 7 and 14 days), following a 3 days exposure to $116 \mu\text{g L}^{-1}$ Roundup®. Both F and p values are given for each variable. Non significant differences are signalized as "ns".

Parameter	Factors					
	Agent		Time		Interaction	
	F	p	F	p	F	p
GDI	152.62	<0.05	83.11	<0.05	31.06	<0.05
GDI_{FPG}	55.02	<0.05	21.44	<0.05	3.03	ns
NSS_{FPG}	3.94	ns	5.48	<0.05	8.47	<0.05
$\text{GDI}_{\text{EndoIII}}$	30.97	<0.05	4.21	<0.05	7.81	<0.05
$\text{NSS}_{\text{EndoIII}}$	2.47	ns	16.25	<0.05	17.48	<0.05

Table 5

Mean frequencies (%) of damaged nucleoid classes ($\pm$ standard error), measured by the comet assay including the incubation with the FPG enzyme in blood cells of *A. anguilla* exposed to $116 \mu\text{g L}^{-1}$ Roundup® for 3 days (Exposure) and after cessation of exposure (1, 7 and 14 days post-exposure). Statistically significant differences ($p < 0.05$) are: (°) in relation to control (C), (♦) in relation to 1 day post-exposure and (▲) in relation to 7 days post-exposure.

			GDI _{FPG} —DNA damage classes—Roundup® experiment				
			0	1	2	3	4
Exposure	3 Days	C	0.00 $\pm$ 0.00	12.25 $\pm$ 1.84	67.50 $\pm$ 1.22	20.25 $\pm$ 0.61	0.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	0.42 $\pm$ 0.30°	36.58 $\pm$ 3.24°	56.92 $\pm$ 0.79°	6.08 $\pm$ 2.21°
Post-exposure	1 Days	C	0.00 $\pm$ 0.00	8.00 $\pm$ 3.27	66.92 $\pm$ 1.29	24.83 $\pm$ 4.76	0.25 $\pm$ 0.20
		R	0.00 $\pm$ 0.00	0.58 $\pm$ 0.36	33.63 $\pm$ 0.83°	59.58 $\pm$ 4.13°	3.63 $\pm$ 0.83°
	7 Days	C	0.00 $\pm$ 0.00	14.75 $\pm$ 0.61	55.83 $\pm$ 0.95	27.42 $\pm$ 1.56	2.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	8.08 $\pm$ 2.29	63.33 $\pm$ 4.67♦	26.92 $\pm$ 6.44♦	1.67 $\pm$ 0.60
	14 Days	C	0.00 $\pm$ 0.00	12.42 $\pm$ 0.88	74.75 $\pm$ 1.02	12.83 $\pm$ 0.14	0.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	10.22 $\pm$ 1.54♦	57.06 $\pm$ 0.47°▲	32.33 $\pm$ 1.96♦	0.39 $\pm$ 0.20♦

different authors, based on different genotoxic endpoints (Çavas and Könen, 2007; Cavalcante et al., 2008; Guilherme et al., 2012b). The participation of oxidative stress on the damage induced by Roundup® was evidenced by the detection of oxidatively altered DNA bases (by means of the enzyme-modified comet assay) both in liver and blood tissues (Guilherme et al., 2012a, 2012b).

In line with the previous findings, the present study confirmed the genotoxic potential of both herbicides, as depicted by the results for 3 days exposure. In addition, the apparent lower involvement of oxidative DNA damage in the action of Garlon® and the association of that damaging mechanism to the action of Roundup® (NSS_{EndoIII} parameter pointed out pyrimidine oxidation) was also confirmed.

The processes that determine DNA damage, and its extent, can be divided according to its intervention towards a genotoxicity promotion and, in opposition, according to its anti-genotoxic role. DNA integrity may be affected as a result of a direct attack of the genotoxicant (e.g. formation of DNA adducts) and/or its metabolites (Bonfanti et al., 1992), as well as, indirectly, through the over-generation of ROS that may disrupt DNA or affect the function of DNA repair enzymes (Shimura-Miura et al., 1999). In addition, a depletion of the overall antioxidant defences caused by the contaminant can create a pro-oxidant condition, which in turn also promotes DNA damage (Ahmad et al., 2006). On the other hand, anti-genotoxic responses may include the increment of DNA repair and antioxidant actions, as well as cell renewal in the target tissue. Keeping in mind the latter process, it should be considered that genetic damage manifestations (e.g. micronuclei, DNA strand breaks) and their detection can be affected by cell renewal (Saleha Banu et al., 2001), which in turn can also be modulated by the genotoxic agent itself (Pacheco and Santos, 2002). Therefore, the occurrence/absence of a significant DNA damage at a given moment depends on the balance between the two types of processes above mentioned. An eventual recovery of damage in a post-exposure period relies on the tendency of the anti-genotoxic processes to gain preponderance in relation to the genotoxic pressures because of the cessation of exposure. Nevertheless, the hypothesis that post-exposure period can be more

critical for manifestation of genotoxicity cannot be overlooked, as demonstrated in fish exposed to a recognized genotoxicant (benzo(a)pyrene) (Rose et al., 2001; Nigro et al., 2002).

The following discussion will be focused on the evolution of damage after removal of the contamination source, as an innovative approach, since it has never been conducted in fish in the context of herbicides genotoxicity.

4.1. Progression of DNA damage induced by Garlon® in post-exposure period

In what concerns to the post-exposure period, it was possible to observe slightly different patterns depending on the parameter considered. Thus, the GDI parameter, although showing a tendency to damage decrease over this period, did not reflect an effective recovery from the damage previously induced, since the control levels were not restored. Reinforcing the weakly supported indication provided by GDI towards recovery, the GDI_{FPG} parameter revealed an actual recovery 14 days after the cessation of exposure, which was substantiated by the damage decreases in relation to days 1 and 7 (also corroborated by the significant effect of the factor time). Differently, GDI_{EndoIII} data cannot be interpreted as reflecting a recovery. Despite the absence of a significant damage increase at day 7 (G vs. C groups), it was not possible to perceive neither a decreasing tendency over the whole period nor a significant decrease in relation to R group at day 1. On the other hand, the analysis of individual classes of damage over the post-exposure period, either considering GDI_{FPG} or GDI_{EndoIII}, displayed a general tendency for the frequency of nucleoids of class 2 to increase, concomitantly to a decrease in class 3, which supports the idea of a reduction of damage extent. Moreover, the interaction agent $\times$ time detected for GDI_{FPG} and GDI_{EndoIII} (two-way ANOVA) did not prevent the confirmation of significant effects for these factors separately.

The submission of fish to a post-exposure period can be regarded as a simulation of the animal migrating from a contaminated to an uncontaminated area. This alteration ensures that the exposure

Table 6

Mean frequencies (%) of damaged nucleoid classes ($\pm$ standard error), measured by the comet assay including the incubation with the EndoIII enzyme in blood cells of *A. anguilla* exposed to $116 \mu\text{g L}^{-1}$ Roundup® for 3 days (Exposure) and after cessation of exposure (1, 7 and 14 days post-exposure). Statistically significant differences ($p < 0.05$) are: (°) in relation to control (C) and (♦) in relation to 1 day post-exposure.

			GDI _{EndoIII} —DNA damage classes—Roundup® experiment				
			0	1	2	3	4
Exposure	3 Days	C	0.00 $\pm$ 0.00	24.25 $\pm$ 1.02	66.75 $\pm$ 0.61	9.00 $\pm$ 0.41	0.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	0.17 $\pm$ 0.17°	33.92 $\pm$ 0.60°	59.67 $\pm$ 0.51°	6.25 $\pm$ 0.25°
Post-exposure	1 Days	C	0.00 $\pm$ 0.00	19.75 $\pm$ 0.20	61.25 $\pm$ 1.84	19.00 $\pm$ 2.04	0.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	0.75 $\pm$ 0.38°	40.42 $\pm$ 4.06	54.75 $\pm$ 4.94°	4.08 $\pm$ 1.31
	7 Days	C	0.00 $\pm$ 0.00	5.00 $\pm$ 0.82	53.25 $\pm$ 7.55	40.42 $\pm$ 7.28	1.33 $\pm$ 1.09
		R	0.00 $\pm$ 0.00	4.83 $\pm$ 0.96	53.92 $\pm$ 3.84	37.42 $\pm$ 3.47	3.83 $\pm$ 1.31
	14 Days	C	0.00 $\pm$ 0.00	14.25 $\pm$ 5.10	72.67 $\pm$ 4.63	13.08 $\pm$ 0.48	0.00 $\pm$ 0.00
		R	0.00 $\pm$ 0.00	7.83 $\pm$ 1.88♦	57.33 $\pm$ 3.18	33.92 $\pm$ 1.96♦	0.92 $\pm$ 0.79

to waterborne contaminants was abolished. However, the xenobiotic(s) previously absorbed can still persist in the organism's tissues, promoting the extension of damage beyond the exposure period. Bearing this in mind, it is important to understand the toxicokinetics and toxicodynamics, mainly of the active ingredient of Garlon® (triclopyr). Usually, the half-life time of a pesticide is defined for its active ingredient. Thus, it is known that triclopyr needs less than a day to be reduced to 50% of its concentration in fish body (Thompson et al., 1991). Considering this, the maintenance of a DNA damaging effect after the cessation of exposure, in the time scale currently assessed, cannot be associated to the persistence of triclopyr in fish's body and the subsequent direct attack to DNA. Hence, the blood cells of eels failed to fully recover from the damage induced by Garlon® reflected, mainly, the incapacity of DNA repair system to restore DNA integrity in a period of 14 days.

A negligible involvement of oxidative damage seems to be depicted in the absence of significant increases of NSS_{FPG} and $NSS_{EndoIII}$ values. However, this assumption should be taken carefully, since, as mentioned above, a recovery of damage value up to the control level was exhibited by GDI_{FPG} on day 14.

4.2. Progression of DNA damage induced by Roundup® in post-exposure period

In what concerns this period, GDI parameter revealed a decrease on the non-specific DNA damage induced by Roundup® after 14 days in herbicide-free water, while GDI_{FPG} and $GDI_{EndoIII}$ failed to express a complete recovery of DNA stability.

In order to understand the recovery pattern evidenced by GDI, it should be clarified if the chemical elimination from the fish body contributed to DNA damage decline. Notably, glyphosate, the active ingredient of Roundup®, has a low potential to bioaccumulate (WHO, 1994) and thus, a substantial depuration (probably achieved 14 days after the cessation of exposure) appears to be an important component of the recovery from damage.

The absence of DNA damage (expressed by GDI) on day 14 could also be explained by the removal of damaged erythrocytes from circulation (hemocatheresis) together with the production of new undamaged cells. The life span of erythrocytes in fish is supposed to be around 100 days, but it can be reduced as a consequence of exposure to contaminants. Increased splenic erythrophagia, indicating an elevated hemocatheresis, was already associated with an intense genetic damage in *A. anguilla* (Pacheco and Santos, 2002). Nevertheless, the overall results do not support increased erythrocyte removal as being important in the apparent recovery from DNA damage, since renewal of the erythrocyte population renovation would also affect GDI_{FPG} and $GDI_{EndoIII}$ scores, which did not happen.

Contrary to what was discussed above for GDI, GDI_{FPG} or $GDI_{EndoIII}$ parameters did not display a complete recovery. It is particularly interesting to that, analysing GDI versus GDI_{FPG} or $GDI_{EndoIII}$, the persistence of damage on day 14 reflected by the latter parameters coincided with the late appearance of DNA breaks corresponding to net enzyme-sensitive sites. NSS_{FPG} and $NSS_{EndoIII}$ indicated the occurrence of oxidative damage on day 14 post-exposure, affecting both purine and pyrimidine bases. These results seem to point out a progressive decay of cell antioxidant protection coupled with the incapacity to repair this particular type of DNA damage.

The lack of a significant effect concerning the factor agent for NSS_{FPG} and $NSS_{EndoIII}$ data in two-way ANOVA reflected the complexity of processes involved, which is corroborated by the interaction agent $\times$ time.

It can be also inferred that, following Roundup® exposure, blood cells of *A. anguilla* appeared to be more successful in repairing

damage detected by GDI (here called non-specific damage), than that associated to base oxidation. However, some repair capacity was still evident when analysing the individual classes of damage, either considering GDI_{FPG} or $GDI_{EndoIII}$, which displayed a general tendency for the frequency of class 3 nucleoids to decrease during the post-exposure period (while class 2 frequency increased).

Therefore, an indirect action of Roundup® on DNA was demonstrated, related to the formation of ROS, occurring in a late phase. This can be due to the redox cycle of Roundup® constituents generating ROS and/or to the inhibition/exhaustion of the antioxidant defences, thus decreasing the ability to counteract a ROS challenge (even at basal levels). Supporting this idea, organophosphate pesticides affect DNA bases either directly or indirectly via protein alkylation (Mohan, 1973). Additionally, and according to Saleha Banu et al. (2001), besides ROS-dependent processes, organophosphate pesticides can also inhibit enzymes involved in DNA repair.

Taking into account the present overall results, the genotoxic endpoints assessed recovered only partially. Even considering the hypothesis that fish can restore their DNA integrity measured by the comet assay in blood cells, a question remains whether a single exposure can lead to long-term genome-destabilizing effects and cause stable genotoxic lesions. The transient nature of chromosomal damage in the context of pesticide exposure has been assessed in human studies. Thus, patients suffering acute organophosphate intoxication evidenced an increase in the frequency of chromosome aberrations, which returned to normal levels 6 months after the acute exposure (van Bao et al., 1974). Another study revealed that the cessation of occupational exposure to phosphine was accompanied by a significant decline in the chromosome rearrangements frequency for one-year time (Garry et al., 1992). Thus, the genotoxic potential of Roundup® currently detected in *A. anguilla* should be regarded as a primary risk factor for long-term effects such as carcinogenesis. In line with this, a late appearance of long-term adverse effects (e.g. carcinogenicity and reproductive impairments) in fish exposed to pesticides has been reported (Kime, 1995; Moore and Waring, 2001; de Campos Ventura et al., 2008).

In an attempt to evaluate the suitability of the adopted genotoxic endpoints on the assessment of genotoxic hazard associated to the tested herbicides in environmentally exposed fish, it was demonstrated that comet assay (standard version and applying the extra step with DNA lesion-specific repair enzymes) applied to blood cells displayed promising features, namely reducing the appearance of false negative results in a time scale of days/weeks in animals that are no longer exposed.

5. Conclusion

The present results confirmed the genotoxicity of the herbicide formulations Garlon® and Roundup® to fish at environmentally realistic concentrations.

Analysing the progression of DNA damage along the post-exposure period (up to 14 days), it was observed that fish exposed to Garlon®, did not achieve a complete recovery, although they showed some tendency to recover. A full recovery from damage was only exhibited by GDI_{FPG} parameter (day 14), pointing to a minor induction of purine base oxidation coupled with an efficient enzymatic repair specifically directed to these DNA lesions.

In relation to Roundup®, a recovery was evident when considering non-specific DNA damage (GDI), on day 14 post-exposure, while GDI_{FPG} and $GDI_{EndoIII}$ failed to express a complete restoration of DNA integrity. In addition, this glyphosate-based herbicide showed a potential to oxidatively damage DNA, depicted mainly as a late (day 14) appearance of DNA breaks corresponding to net enzyme-sensitive sites (NSS_{FPG} and $NSS_{EndoIII}$). Moreover, blood cells of *A.*

anguilla exposed to Roundup® appeared to be more successful in repairing damage detected by GDI (non-specific damage) than that associated with base oxidation.

Overall, the present findings highlight the genetic hazard to fish associated to the addressed agrochemicals, reinforcing the hypothesis of long-lasting damage.

Statement of authors contribution

Drs Sofia Guilherme, Isabel Gaivão, Maria Ana Santos and Mário Pacheco planned the experimental design. Dr Sofia Guilherme performed the practical component and prepared the manuscript draft with important intellectual input from Drs Isabel Gaivão, Maria Ana Santos and Mário Pacheco. All authors approved the final manuscript.

Ethical statement

This study was conducted in accordance with the EU Directive 2010/63/EU on the protection of animals used for scientific purposes, under the supervision of a team member (Mário Pacheco) authorized by the competent authorities.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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